PRANJAL SINGH

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PROFESSIONAL SUMMARY

Final-year BCA (Artificial Intelligence & Data Science) student and full-stack developer with hands-on experience across two internships, building REST APIs, AI-integrated features, and production-deployed web apps using React.js, Node.js, Express.js, and MongoDB. Built a relational database engine from scratch (StructDB) with B-Tree indexing, hash-table lookups, and full ACID transaction support. Strong foundations in Data Structures & Algorithms, OOP, and DBMS from building StructDB's core engine. Seeking Software Engineer / Full-Stack Developer / Associate Engineer roles.

EDUCATION

Graphic Era Hill University — B.C.A., Artificial Intelligence & Data Science **Jun 2024 – Jun 2027**

Bhimtal, Uttarakhand | CGPA: 7.82 / 10.0

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C, C++, SQL

Frontend: React.js, Next.js, Vite, HTML5, CSS3, Responsive Web Design

Backend: Node.js, Express.js, Flask, RESTful API Development, JWT Authentication, Bcrypt, Socket.io, Multer

Databases: MongoDB (Mongoose), MySQL, SQLite, PostgreSQL — Relational & NoSQL Database Design

Tools & Platforms: Git, GitHub, Postman, Vercel, Render, MongoDB Atlas, Linux, VS Code, CI/CD

Core CS: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Agile/Scrum, Role-Based Access Control (RBAC)

INTERNSHIP EXPERIENCE

Full Stack Developer Intern — Technology Business Incubator (TBI), GEHU **Jun 2026 – Aug 2026**

Project: AgriConnect | React.js, Next.js, Node.js, Express.js, MongoDB, Google Gemini API

Live Demo

- Built REST APIs for an AI-powered farmer–buyer marketplace, including product listings, buyer-seller matching, and a dedicated crop-advisory endpoint (POST /api/ai/crop-advice) integrating Google Gemini 2.5 Flash for real-time diagnosis and treatment suggestions.
- Designed MongoDB/Mongoose schemas with validation for agricultural data; implemented JWT authentication, bcrypt password hashing, and rate limiting across protected routes; followed Agile/Scrum with Git-based version control.
- Built the frontend AI Advisor interface with structured output cards, loading states, and user-friendly error handling; documented the prompt-engineering approach for consistent JSON output.

Backend Developer Intern — Uptoskills (Remote) **May 2026 – Aug 2026**

Project: Learning Management System | Node.js, Express.js, MongoDB, JWT

- Built and integrated an Interview Management Module and dashboard into an existing LMS backend, designing 6+ REST API endpoints for interview scheduling, candidate records, and dashboard data with full CRUD operations, input validation middleware, and JWT authentication.
- Audited 20+ existing API routes across the LMS codebase; identified and fixed 8 broken endpoints, resolving issues with request handling, middleware conflicts, and data validation.

PROJECTS

MyRideShare — AI-Powered Ride Sharing Application **2025**

React.js · Node.js · Express.js · MongoDB · Socket.io · OpenRouteService API · Google Gemini AI

GitHub | Live Demo

- Built a full-stack ride-sharing platform with separate User and Captain portals; real-time ride booking and live location tracking via Socket.io WebSockets.
- Integrated OpenRouteService API for route mapping, distance calculation, and dynamic fare estimation across three vehicle types; used Google Gemini AI for location recommendations.
- Secured with JWT authentication and bcrypt password hashing; protected API routes for both users and captains.

StructDB — Custom Relational Database Engine **Nov 2024 – Jan 2025**

Python · Custom SQL Parser · B-Tree · Hash Table · SHA-256 · Tkinter

GitHub

- Built a relational DBMS from scratch with a custom SQL parser (DDL/DML); B-Tree indexing for O(log n) range queries and a hash table for O(1) lookups — zero third-party libraries.
- Implemented full ACID transaction support (BEGIN, COMMIT, ROLLBACK); per-user database isolation, SHA-256 password hashing, and a Tkinter GUI with syntax highlighting and an analytics dashboard.

Price Comparison Pro — Full Stack Web App **Jan 2025 – Apr 2025**

Python · Flask · React.js · Vite · SQLite · JWT · BeautifulSoup4

GitHub | Live Demo

- Scraped real-time prices across Amazon, Flipkart, Ajio, and Meesho using a parallel ThreadPoolExecutor; built a Flask REST API with server-side caching, JWT authentication, and per-user search history.

CERTIFICATIONS

Developing Soft Skills — NPTEL | PHP & MySQL — IIT Bombay

Python for Data Science — Saylor Academy | C++ Programming — Saylor Academy